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8.3 Invariants — Examples

The loop invariant is true in four crucial points in a loop. Using the loop invariant, we can construct the loop and reason about the properties of the variables at these points.

Example 8.5. Design an iterative algorithm to compute an. Let us name the algorithm power(a, n). For example,

power(10, 4) = 10000

power (5 , 3) = 125

power (2 , 5) = 32

Algorithm power(a, n) computes a^n by multiplying a cumulatively n times.

$$\mathbf{a^n = \underbrace{ax\ ax\ \ldots\ x\ a}_{\text{n times}}}$$

The specification and the loop invariant are shown as comments.

```
power (a, n)
-- inputs: n is a positive integer
-- outputs: p = a^n
p, i := 1 , 0
while i ≠ n
   -- loop invariant: p = a^i
   p, i :=p X a, i+1
```

The step by step execution of power (2, 5) is shown in Table 8.1. Each row shows the values of the two variables p and i at the end of an iteration, and how they are calculated. We see that $p = a^i$ is true at the start of the loop, and remains true in each row. Therefore, it is a loop invariant.

iteration	p	p x a	i	i+1	a^i
0	1		0		2^0
1	2	1x2	1	0 + 1	2^1
2	4	2x2	2	1 + 1	2^2
3	8	4x2	3	2 + 1	2^3
4	16	8x2	4	3 + 1	2^4
5	32	16x2	5	4+1	2^5

Table 8.1: Trace of power (2, 5)

When the loop ends, $p = a^1$ is still true, but i = 5. Therefore, $p = a^5$. In general, when the loop ends, $p = a^n$. Thus, we have verified that power(a, n) satisfies its specification.

Example 8.6. Recall the Chocolate bar problem of Example 1.11. How many cuts are needed to break the bar into its individual squares?

We decided to represent the number of pieces and the number of cuts by variables p and c respectively. Whenever a cut is made, the number of cuts increases by one and the number of pieces also increases by one. We decided to model it by an assignment.

p, c := p + 1, c+1

The process of cutting the bar can be modeled by a loop. We start with one piece and zero cuts, p = 1 and c = 0. Let n be the number of individual squares. When the number of pieces p equals the number